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COPIA FIEL DE CERTIFICADOS

Producto MINIT TEXTILES



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Marco de Reconocimiento Internacional (Reciprocidad)

El sistema de aceptación de ensayos extranjeros en Argentina se basa en acuerdos internacionales de reciprocidad que eliminan la necesidad de repetir pruebas locales. Recientemente, el marco normativo se ha simplificado mediante el Decreto 892/2025.

El Gobierno argentino estableció el reconocimiento automático de certificaciones y ensayos internacionales para agilizar el comercio:

- Se consideran válidos los ensayos realizados en países con altos estándares de vigilancia técnica (países de "alta vigilancia") detallados en la normativa.
- Los productos que posean certificaciones de laboratorios acreditados por entidades avaladas por acuerdos de reciprocidad no requieren nuevos ensayos locales.

Principales Organismos de Acreditación y Cooperación Regional e Internacional

Organismo	Alcance	Rol Principal
ILAC	Global	Cooperación para laboratorios de ensayo y calibración.
IAF	Global	Reconocimiento de organismos de certificación de productos y sistemas.
IAAC	Regional (Américas)	Cooperación Interamericana de Acreditación.
OCDE	Global	Acuerdos de "Aceptación Mutua de Datos" (MAD) para seguridad ambiental y sanitaria.
A2LA	Regional (Américas)	Asociación para la Acreditación de Laboratorios. Bajo la norma ISO/IEC 17025. Garantiza la competencia técnica y la confianza global.
ANAB / ANSI	Regional (Américas)	Junta de Acreditación. Organismo de acreditación de normas, laboratorios y organismos de inspección y certificación (ISO/IEC 17025, 17020, 17021, 17034).

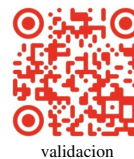


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Producto MINIT TEXTILES



INTI-CIT-IRAM G7577 (INFLAMABILIDAD VERTICAL DE TEXTILES NIVEL 1 – CLASE A)

INTI  Construcciones

INFORME DE ENSAYO

Solicitante: MINOTTI PABLO DAVID

O.T.: 101/20631

Pág.: 1 de 2

Fecha: 03/05/2011

Informe: Único.

Dirección: Cochrane 2322

(1419) – Cdad. Autónoma de Buenos Aires

1. OBJETIVO

Determinación de la **Inflamabilidad Vertical de Textiles**

2. MATERIAL

Una (1) muestra de tela color blanca, de composición no declarada, identificada por el solicitante como: “Tela con tratamiento retardante marca MINIT de Pablo Minotti”

Nota: La tela presenta una textura acartonada como resultado del tratamiento aplicado, no habiendo sido informado el proceso de aplicación del mismo.



3. MÉTODO EMPLEADO

El ensayo de **Inflamabilidad Vertical de Textiles** se realizó de acuerdo a las indicaciones de la Norma IRAM-INTI-CIT G 7577, “MATERIALES TEXTILES – Método de ensayo de comportamiento a la llama con la probeta vertical”.

La muestra fue recibida el día 26/04/2011 y se ensayó el día 02/05/2011

4. RESULTADOS OBTENIDOS

La clasificación se hace teniendo en cuenta las referencias consignadas a continuación:

Análisis cualitativo:

- Entra en ignición y la llama se propaga.
- Entra en ignición, pero es autoextinguible.
- No entra en ignición.
- Funde
- Se contrae y se aleja de la llama.
- Gotea y caen trozos encendidos.

Este informe no podrá ser reproducido parcialmente sin la autorización escrita del Laboratorio. Los resultados consignados se refieren exclusivamente a los elementos recibidos, el INTI y su Centro de Investigación y Desarrollo en Construcciones declinan toda responsabilidad por el uso indebido o incorrecto que se hiciera de este informe.

Instituto Nacional de Tecnología Industrial
Centro de Investigación y Desarrollo
en Construcciones

Avenida General Paz 5445
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Teléfono (54 11) 4724 6200
e-mail: construcciones@inti.gob.ar



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Solicitante: MINOTTI PABLO DAVID

Dirección: Cochrane 2322
(1419) – Cdad. Autónoma de Buenos Aires

O.T.: 101/20631
Pág.: 2 de 2
Fecha: 03/05/2011
Informe: Único.

Análisis cuantitativo:

TIEMPO DE LLAMA:

- NIVEL 1: Menor o igual a 1 segundo.
- NIVEL 2: Mayor a 1 segundo, pero menor o igual a 5 segundos.
- NIVEL 3: Mayor a 5 segundos.

TIEMPO DE INCANDESCENCIA:

- NIVEL 1: Menor o igual a 3 segundos.
- NIVEL 2: Mayor a 3 segundos, pero menor o igual a 15 segundos.
- NIVEL 3: Mayor que 15 segundos.

LONGITUD DAÑADA:

- NIVEL 1: Menor o igual a 100 milímetros.
- NIVEL 2: Mayor a 100 milímetros, pero menor o igual a 150 milímetros.
- NIVEL 3: Mayor a 150 milímetros.

GOTEO:

- NIVEL 1: Ausencia de goteo.
- NIVEL 2: Gotas y/o partes desprendidas que se apagan en el momento de contacto con el piso de la cabina.
- NIVEL 3: Gotas y/o partes desprendidas que siguen encendidas en el piso de la cabina menos de 3 segundos.

Se clasifica en 3 niveles, siendo 1 el mejor y 3 el peor.

“Tela con tratamiento retardante”

- Análisis cualitativo:

No entra en ignición.

- Análisis cuantitativo:

Sentido de la muestra	Tiempo de duración de la llama (s)	Tiempo de incandescencia (s)	Longitud carbonizada (mm)
Longitudinal	0	1	36
Transversal	0	1	27

De acuerdo a los resultados obtenidos, el material: muestra de tela color blanca identificada por el solicitante como “Tela con tratamiento retardante” clasifica como **NIVEL 1.**

Arq. BASILIO MASAPON
COORDINADOR
U.T. TECNOLOGIA EN INCENDIOS
INTI-CONSTRUCCIONES

Ing. VICENTE L. VOLANTINO
DIRECCION
INTI-CONSTRUCCIONES

Nota:

De acuerdo a reglamentaciones internacionales, estos ensayos deben considerarse para medir y describir el comportamiento del material bajo condiciones controladas pero no se puede estimar cuál será el comportamiento del mismo si se modifican total o parcialmente las condiciones de ensayo.

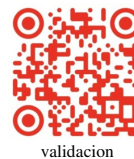


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NFPA 701 (INFLAMABILIDAD VERTICAL DE TEXTILES NIVEL 1 – CLASE A)

INTI Incendios y Explosiones



Presidencia
de la Nación

Ministerio de
Producción

Informe de Ensayo

OT N° 131- 161
Informe: Único
Página 1 de 3

Fecha de Informe: 30/08/2016

Solicitante	Minotti Pablo David Echeverría 2073 – Piso 4 – Dto. B
Elemento	Textil negro
Determinaciones requeridas	Evaluación de la Inflamabilidad Vertical de Textiles
Fecha de Recepción	13 de julio de 2016
Fecha de ensayo	24 de julio de 2016

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Teléfono (54 11) 4724 6200
E-mail incendiosyexplosiones@inti.gob.ar

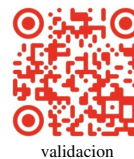


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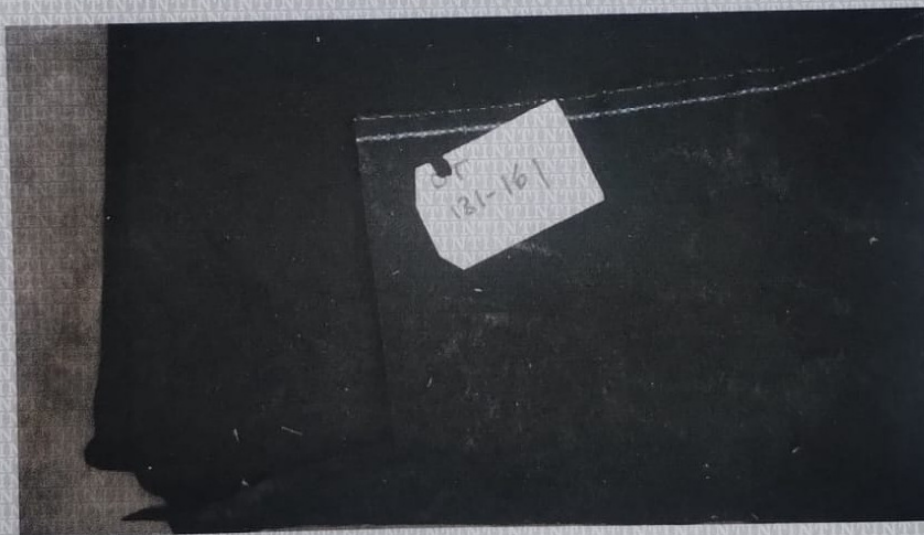
OT N° 131-161
Informe: Único
Página 2 de 3

1. METODOLOGÍA EMPLEADA

El ensayo se realizó de acuerdo a las indicaciones de la Norma NFPA 701, "Standard Methods of Fire Tests of Flame Propagation of Textiles and Films: Test Method 1".

2. IDENTIFICACIÓN DE LA MUESTRA

Textil negro identificado por el solicitante como "TEXTIL NEGRO SINTÉTICO CON TRATAMIENTO RETARDANTE IGNIFUGO MINIT"



Nota: La muestra presentaba partículas salinas sobre la superficie que se desprendían durante su manipulación.



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Producción

OT N° 131-161
Informe: Único
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3. RESULTADOS OBTENIDOS

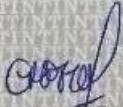
- Análisis cuantitativo:

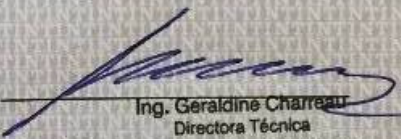
Probeta	Peso inicial (g)	Peso final (g)	Tiempo de llama (seg)	Tiempo de llama material desprendido (seg)	Porcentaje de masa perdida (%)
1	35,21	34,07	0	0	3,24
2	33,3	31,29	0	0	6,04
3	41,68	39,73	0	0	4,68
4	34,28	32,06	0	0	6,48
5	38,09	36,5	0	0	4,17
6	35,5	33	0	0	7,04
7	40,36	38,5	0	0	4,61
8	39,21	37,01	0	0	5,61
9	39,24	37,5	0	0	4,43
10	39,1	34,9	0	0	10,74
Promedio	37,597	35,456	0	0	5,70
Desvio Standard					2,00

De acuerdo a los resultados obtenidos, el material identificado por el solicitante como:
"TEXTIL NEGRO SINTÉTICO CON TRATAMIENTO RETARDANTE IGNIFUGO MINIT"
CUMPLE con los requerimientos de la norma NFPA 701:1999 Standard Methods
of Fire Test for Flame Propagation of Textiles and Films -Test method 1: "Curtains,
draperies, or other windows treatments".

Observaciones:

De acuerdo a reglamentaciones internacionales, estos ensayos deben considerarse para medir y describir el comportamiento del material bajo condiciones controladas, pero no se puede estimar cuál será el comportamiento del mismo si se modifican total o parcialmente las condiciones de ensayo.


Esp. Arq. M. Paula Cheheid
INTI - Incendios y Explosiones


Ing. Geraldine Charreaur
Directora Técnica
INTI - Incendios y Explosiones

Fin del Informe



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NFPA 705 (INFLAMABILIDAD VERTICAL DE TEXTILES NATURALES NIVEL 1 – CLASE A) Govmark/SGS (<https://www.sgs.com/en/service-groups/fire-testing>) Convenio de reciprocidad INTI / SGS, 23/08/2001



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Farmingdale, New York 11735-5626 USA
Tel. +1 (631) 293-8944 Fax +1 (631) 293-8956
e-mail: testing@govmark.com

Page 1

Received:04/20/2016	Completed:04/20/2016	Letter: M	MM	P.O.#:	Test Report #:	3-12898-0-
Client's Identification	Style: Fabric. Content: 100% Cotton. Additional Information: 100% Cotton Fabric Treated by Spray with Minit Flame Retardante for Fabrics.					
Tested For: Pablo Minotti Flame Retardant Technologies LLC 1330 NW 78th Ave Miami, FL 33126			Key Test: NFPA 705 225 Tel: 1-(305)-400-8889 Fax: 1-()- -			
LE 2013; V 09/15 ; NE 2018 TEST PERFORMED: NFPA 705 - Recommended Practice for a Field Flame Test for Textiles and Films (2013 Edition) 1.2.1 The purpose of this recommended practice is to provide authorities having jurisdiction with a field means of determining the tendency of textiles and films to sustain burning subsequent to the application of a relatively small, open flame. 1.2.2: The methods described herein and the results do not correlate with any known test method, and factors relating to reproducibility and correlation have not been determined; therefore, they should not be relied upon when more definitive test data are available. 1.3.1.2: The field test method has utility only when the authority having jurisdiction has no reliable data and, therefore, is forced to rely solely on the field test findings. 1.3.2: Materials applied to surface of buildings or backing materials as interior finishes in buildings should be tested and classified in accordance with ASTM E 84 or NFPA 286. In the case of textile wall coverings, the use of NFPA 265 is also appropriate. * 5.1 Limitations: The deficiencies and limitations of the field test method can lead to misleading or erroneous results, and the error can be in both directions. It is quite possible to have a too-small sample show several seconds of afterflaming, causing the material to be rejected. It is equally possible for improper or inadequate field procedures to incorrectly indicate satisfactory flame resistance. This can result in dangerous errors. 5.2 Precautions: Field procedures are useful, but they must be used with good judgment and their limitations should be recognized. Field tests should not be relied on as the sole means for ensuring adequate flame resistance of decorative materials. They are, however, useful in augmenting a comprehensive regulatory program. BRIEF DESCRIPTION OF TEST (Chapter 4 Procedure): The test should be performed in a draft-free and safe location free of other combustibles. Specimens should be removed from the existing material. Specimens should be dry and should be a minimum of 12.7 mm x 101.6 mm (0.5" x 4.0"). The fire exposure should be from a common wood match or source with equivalent flame properties. The sample should be suspended (preferably by means of a spring clip, tongs, or similar device) with the long axis vertical, the flame applied to the center of the bottom edge and the bottom edge 12.7 mm (0.5") above the bottom of the flame. After 12 seconds of exposure the match is to be removed gently away from the sample. (Page 1 of 3)						

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e-mail: testing@govmark.com

Page 2

Received: 04/20/2016	Completed: 04/20/2016	Letter: M	MM	P.O.#:	Test Report #:	3-12898-0-
Client's Identification	Style: Fabric. Content: 100% Cotton. Additional Information: 100% Cotton Fabric Treated by Spray with Minit Flame Retardante for Fabrics.					
Tested For: Pablo Minotti			Key Test: NFPA 705		225	
Flame Retardant Technologies LLC 1330 NW 78th Ave Miami, FL 33126			Tel: 1-(305)-400-8889 Fax: 1-()- -		Ext:	
TEST LOCATION: ☒ On-site ☐ Govmark Laboratory **						
FLOOR-TO-BOTTOM-OF-SPECIMEN HEIGHT: ☒ 8 inches ☐ Other _____ inches						
RESULTS:						
Direction	Specimen #	Ignition Observed (yes/no)	Afterflame (seconds)	Drip Burn (seconds)	Flaming (inches)	
MD	1	Yes	0	0	0.2	
CMD	2	Yes	0	0	0.3	
Note 1: The "Ignition Observed" column information is an observation only and does not enter into the acceptance criteria.						
Note 2: NFPA 705 does not specify the number of specimens to be tested.						
ABBREVIATIONS: MD = Machine (Warp) ; CMD = Cross Machine (Fill/Weft) ; CL = Complete length burn						
ACCEPTANCE CRITERIA***:						
Afterflame: There should be no more than 2.0 seconds of afterflame for any individual specimen.						
Drip Burn: Materials that break or drip flaming particles should be rejected if the materials continue to burn after they reach the floor****						
Flaming: During the exposure, flaming should not spread over the complete length of the sample or, in the case of larger samples, in excess of 4 inches from the bottom of the sample.						
REMARKS: None.						
CONCLUSION: Based on the above Results and Acceptance Criteria, the item tested:						
☒ Complies; ☐ Does not comply						
CERTIFICATION: I certify that the above results were obtained after testing specimens in accordance with the procedures and equipment specified by NFPA 705.						
AUTHORIZED SIGNATURE GOVMARK /pm			APR 26 2016 <i>Robert I. Brown</i>			
			(Page 2 of 3)			

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Page 3

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Client's Identification	Style: Fabric. Content: 100% Cotton. Additional Information: 100% Cotton Fabric Treated by Spray with Minit Flame Retardante for Fabrics.					
Tested For: Pablo Minotti				Key Test: NFPA 705		225
Flame Retardant Technologies LLC				Tel: 1-(305)-400-8889		Ext:
1330 NW 78th Ave				Fax: 1-()- -		
Miami, FL 33126						

GOVMARK ADDENDUM:

* In addition to the NFPA listings, other furnishings tests might apply, such as NFPA 260, NFPA 261, and California Technical Bulletin 133 for upholstery products, and NFPA 701 Test Method 1 for drapery type products, or any other code specified furnishings tests.

** While NFPA 705 is an on-site test some client's, either on their own or at the request of a code official, have opted to request performance of this test at the Govmark laboratory facility.

DISCUSSION:

*** The "ACCEPTANCE CRITERIA" is adopted from the "Requirements" section of NFPA 705 Chapter 4 Procedure. Where the word "should" appears in the "Requirements" section, it may fall under its definition per NFPA 705 paragraph 3.2.3, which states:

3.2.3 Should. Indicates a recommendation or that which is advised but not required.

It is Govmark's decision to take these recommendations under advisement, adopt them as the requirement, and issue a conclusion based upon them.

**** NFPA 705 does not specify a "floor-to-bottom-of-specimen" height from which the "Drip Burn" criteria can be measured. Unless specified otherwise by the client, when testing at Govmark, a "floor-to-bottom-of-specimen" height of 8 inches is utilized. During field testing, the height may be adjusted accordingly to be representative of the installation (i.e. drapery) under test.

(Page 3 of 3)

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CAL 117 – CALIFORNIA MARSHAL (INFLAMABILIDAD HORIZONTAL DE TEXTILES NATURALES = PASS) HOMOLOGACIÓN CON IRAM 11910. RESULTADO equivalente RE 1



Quality Assurance & Compliance Testing
Utilizing Textile & Related Technologies

19 West 36th Street, 10th Floor
New York, NY 10018
Tel: 212 947 8391 Fax: 212 947 8719

www.vartest.com

ISO/IEC 17025 Certified Third Party Test Report

FILE: FLAMER.A0329168
SAMPLE IDENTIFIED BY CLIENT AS:
Fabric Submitted
100% Cotton
Color White

CALIFORNIA TECHNICAL BULLETIN 117-2013 SECTION 1 COVER FABRIC TEST

	<u>CHAR LENGTH (mm)</u>
Specimen #1:	12
Specimen #2:	12
Specimen #3:	10

- a) The mock-up test specimen continues to smolder after the 45-minute test duration: **NO**
- b) A char develops more than 1.8 inches (45 mm) in any direction from the cigarette on the cover fabric measured from its nearest point: **NO**
- c) The mock-up test specimen transitions to open flaming: **NO**

RESULTS: Pass

A material is considered to pass or fail based on the following criteria:

1. A single mock-up test specimen fails to meet the requirements of this test procedure if any of the following criteria occurs:
 - a) The mock-up test specimen continues to smolder after the 45-minute test duration.
 - b) A char develops more than 1.8 inches (45 mm) in any direction from the cigarette on the cover fabric measured from its nearest point.
 - c) The mock-up test specimen transitions to open flaming.
2. The cover fabric passes the test if three initial mock-up specimens pass the test, i.e., the cigarettes burn their full length and the mock-ups are no longer smoldering.
3. If more than one initial specimen fails, the cover fabric fails the test.
4. If any one of the three initial specimens fails, repeat the test on an additional three specimens.
5. If all three additional specimens pass the test, the cover fabric passes the test. If any one of the three additional specimens fails, the cover fabric fails the test.

Signed For The Company By

Adam R. Vdrley
Technical Director

JG/03/649



Stacy Sadowy
Stacy Sadowy
Quality Assurance Supervisor

The findings and results in this test report apply only to the specific sample(s) submitted to us by the client for testing.





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CAL 117 – CALIFORNIA MARSHAL (INFLAMABILIDAD HORIZONTAL DE TEXTILES SINTETICOS = PASS) HOMOLOGACIÓN CON IRAM 11910. RESULTADO equivalente RE 1



Quality Assurance & Compliance Testing
Utilizing Textile & Related Technologies
19 West 36th Street, 10th Floor
New York, NY 10018
Tel: 212 947 8391 Fax: 212 947 8719
www.vartest.com

ISO/IEC 17025 Certified Third Party Test Report

FILE: FLAMER.A032916A
SAMPLE IDENTIFIED BY CLIENT AS:
Fabric Submitted
100% Polyester
Color Green

CALIFORNIA TECHNICAL BULLETIN 117-2013 SECTION 1 COVER FABRIC TEST

	<u>CHAR LENGTH (mm)</u>
Specimen #1:	18
Specimen #2:	15
Specimen #3:	20

- a) The mock-up test specimen continues to smolder after the 45-minute test duration: **NO**
- b) A char develops more than 1.8 inches (45 mm) in any direction from the cigarette on the cover fabric measured from its nearest point: **NO**
- c) The mock-up test specimen transitions to open flaming: **NO**

RESULTS: Pass

A material is considered to pass or fail based on the following criteria:

1. A single mock-up test specimen fails to meet the requirements of this test procedure if any of the following criteria occurs:
 - a) The mock-up test specimen continues to smolder after the 45-minute test duration.
 - b) A char develops more than 1.8 inches (45 mm) in any direction from the cigarette on the cover fabric measured from its nearest point.
 - c) The mock-up test specimen transitions to open flaming.
2. The cover fabric passes the test if three initial mock-up specimens pass the test, i.e., the cigarettes burn their full length and the mock-ups are no longer smoldering.
3. If more than one initial specimen fails, the cover fabric fails the test.
4. If any one of the three initial specimens fails, repeat the test on an additional three specimens.
5. If all three additional specimens pass the test, the cover fabric passes the test. If any one of the three additional specimens fails, the cover fabric fails the test.

Signed For The Company By

Adam R. Varley
Adam R. Varley
Technical Director

JG/03/648



Stacy Sadowy
Stacy Sadowy
Quality Assurance Supervisor

The findings and results in this test report apply only to the specific sample(s) submitted to us by the client for testing.





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minitargentina@gmail.com www.minitfireretardant.com

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validacion

BS 5852: 2006, & CAL TB 117: 2013 PASS (INFLAMABILIDAD HORIZONTAL EN ALFOMBRAS SINTETICAS) HOMOLOGACIÓN CON IRAM 11910. RESULTADO equivalente RE 1



Industrial Global Standards LLC
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TEST REPORT

Nº 1143-72-144

LAB SERVICE

VALIDO SOLO CON CERTIFICADO DE COMPRA.
MINIT IGNIFUGOS

Applicant/ Manufacturer	MINIT Fire Retardant (MINIT Ignifugos) Pablo David Minotti ID: 20224196270 Argentina
Product	Synthetic Carpet treated with MINIT Ignifugos Fire Retardant
Reference Standard/ Method Tests	BS 5852 : 2006, & CAL TB 117 : 2013
Description of Test Specimen	Synthetic carpet (provided by Applicant/ Manufacturer) - 1 piece treated with MINIT ignifugos fire retardant Size of Specimen : 1500mm x 1500mm Nominal Thickness : 3mm
Date Received	May 15, 2025
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Pages	2

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Following the treatment described above, the material was conditioned and tested according to **BS 5852 : 2006, & CAL TB 117 : 2013 Methods of Test for the Assessment of the Ignitability of Upholstered Seating by Smouldering and Flaming Ignition Sources, Clause 11, using Ignition Source 5.**

The results relate only to the ignitability of the combinations of upholstery composites (Clause 11) under the particular condition of test stated; they are not intended as a means of assessing the full potential fire hazard of the item in use.

RESULTS	
Specimen number	1
Duration of flaming of ignition source (min)	3'24"
Duration of flames (min)	3'47"
Duration of smoke (min)	4'19"
Extent of damage of horizontal component	
- width (mm)	123
- length (mm)	121
- depth (mm)	23
Extent of damage of vertical component	
- width (mm)	157
- depth (mm)	27
Progressive smouldering >100mm from ignition source	No
Result	 Pass

CONCLUSION

The specimen showed non-ignition, BS 5852: 2006, Rating NIS. & CAL TB 117: 2013, This Rating represents a Pass for Crib 5.

END OF REPORT



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Homologación con normas IRAM Cal TB117

Norma USA	Ensayo	Norma IRAM equivalente / comparable	Observación
TB 117 / TB 117-2013	Inflamabilidad horizontal de textiles y tapizados	IRAM 11910-3	Evaluación de reacción al fuego de materiales textiles
TB 117 sección horizontal	Velocidad de propagación superficial de llama	IRAM 3904	Similar criterio de propagación superficial
California TB117	Encendido superficial y propagación	IRAM-IAS U 500-250 (aplicaciones textiles)	Uso frecuente en homologaciones locales
ASTM D5132 / UFAC asociados	Smoldering / cigarrillo	IRAM basadas en ISO 8191	Tapizados y mobiliario

“El ensayo California Technical Bulletin TB117 presenta criterios equivalentes de evaluación de propagación superficial de llama horizontal y comportamiento frente a ignición inicial comparables con los establecidos en IRAM 11910 e IRAM 3904 para materiales textiles y revestimientos.”

TB117 (California)

- Ensayo orientado a:
 - muebles tapizados
 - espumas
 - textiles decorativos
- Evaluación horizontal
- Fuente de ignición pequeña
- Uso muy extendido en USA

IRAM 11910

- Más utilizada en:
 - construcción
 - revestimientos
 - cortinados
 - textiles arquitectónicos
- Clasificación de reacción al fuego
- Compatible conceptualmente con ISO 9705 / ISO 5660 según aplicación

El método California TB117 de inflamabilidad horizontal de textiles presenta correlación técnica con los ensayos de reacción al fuego contemplados en IRAM 11910 para materiales textiles y revestimientos interiores, dado que ambos evalúan la propagación superficial de llama y comportamiento frente a fuentes de ignición de baja energía.



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ENSAYOS DE EMANACIÓN DE HUMO Y GASES



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TEST REPORT

Nº 1143-27-522

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VÁLIDO SOLO CON CERTIFICADO DE COMPRA.

MINIT IGNIFUGOS

Applicant/ Manufacturer	MINIT Fire Retatadant (MINIT Ignifugos) Pablo David Minotti ID: 20224196270 Argentina
Product	MINIT Ignifugos / Fire retardant applied on different specimens described below
Reference Standard/ Method Tests	BSS 7239-88. // ASTM E-662. // EN 13823. // ISO 5659-2. TESTS METHODS FOR TOXIC GAS GENERATION BY MATERIALS ON COMBUSTION.
Description of Test Specimens	REF A -Total pieces with fire retardant (provided by Applicant/ Manufacturer) - 4 piece of 100% cotton fabric, treated with MINIT Ignifugos/ Fire retardant - Ref 1 - 4 piece of 100% polyester fabric, treated with MINIT Ignifugos/ Fire retardant - Ref 2 - 4 piece of synthetic carpet, treated with MINIT Ignifugos/ Fire retardant - Ref 3 - 4 piece of synthetic grass carpet, treated with MINIT Ignifugos/ Fire retardant - Ref 4 - 4 piece of foam, treated with MINIT Ignifugos/ Fire retardant - Ref 5 - 4 piece of wood, treated with MINIT Ignifugos/ Fire retardant - Ref 6 REF B -Total pieces without fire retardant (provided by Applicant/ Manufacturer) - 2 piece of 100% cotton fabric, without fire retardant - Ref 1 - 2 piece of 100% polyester fabric, without fire retardant - Ref 2 - 2 piece of synthetic carpet, without fire retardant - Ref 3 - 2 piece of synthetic grass carpet, without fire retardant - Ref 4 - 2 piece of foam, without fire retardant - Ref 5 - 2 piece of wood, without fire retardant - Ref 6 Size of Specimens: 3" x 3" , specified for test methods Size of Specimens: 15" x 15" , specified for test methods
Date Received	May 20, 2025
Issued Date	Jun 27, 2025
Pages	3

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BSS 7239-88 is a Boeing standard test method that measures the concentration of toxic gases produced during the combustion of materials, specifically focusing on those used in aircraft interiors. It is a critical tool for evaluating fire safety, especially in the confined spaces of an aircraft cabin, by assessing the potential hazards from toxic gases like carbon monoxide, hydrogen fluoride, hydrogen cyanide, hydrogen chloride, and sulfur dioxide. The test is performed in a controlled environment, where a material sample is burned and the resulting gases are analyzed.

ASTM E-662 is a standard test method that measures the amount of smoke produced by solid materials when exposed to heat and flame, specifically focusing on the specific optical density of the smoke. It's used to evaluate how much light is blocked by the smoke generated by a material, which is an indicator of visibility reduction in a fire. This test is crucial for assessing fire hazards and is often a requirement in various industries and transportation codes. The test measures the "specific optical density," which is a normalized value that accounts for the amount of smoke generated relative to the sample's exposed area and volume.

EN 13823, is a Single Burning Item (SBI) test, is a European standard for evaluating the fire behavior of building products (excluding flooring) when exposed to a thermal attack. The product is mounted in a corner configuration with two walls (or wings) forming a 90-degree angle. A propane gas burner (30kW) is placed at the bottom of the corner to simulate a fire source. The specimen and burner are placed inside an enclosure with an exhaust system that collects combustion gases. Instruments measure heat release, smoke production, and flame spread, while visual observations record burning droplets and other relevant characteristics. The SBI test results are used to classify products into Euroclasses (A1, A2, B, C, D, etc.) which indicate their fire performance.

ISO 5659-2 is an international standard that specifies a method for measuring smoke production from materials, particularly plastics or synthetic, when exposed to heat and flame. It focuses on determining the optical density of the smoke produced in a single-chamber test. This standard is primarily used for research and development, as well as in fire safety engineering for buildings, trains, and ships, rather than for setting building code ratings.



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TEST REPORT

Nº 1143-27-522

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BSS 7239-88 RESULTS:

Relative Room Humidity: 20% / Test Duration: 20 min. // Chamber Wall Temp: 35°C

Carbon Monoxide (CO ppm) at 1.5 minutes
at 4.0 minutes
at maximum

Nitrogen Oxides (as NO₂ ppm)
Sulfur Dioxide (SO₂ ppm)
Hydrogen Chloride (HCl ppm)
Hydrogen Fluoride (HF ppm)
Hydrogen Cyanide (HCN ppm)

REF A	HCl	HCN	HF	NO ₂	SO ₂	CO 1.5	CO 4.0	CO MAX
1	< 1	ND	ND	< 1	ND	5	55	118
2	< 2	ND	ND	< 1	ND	8	98	211
3	< 3	ND	< 2	< 1	< 1	9	101	218
4	< 3	ND	< 2	< 1	< 1	9	125	238
5	< 3	ND	< 2	< 1	< 1	9	127	247
6	< 2	ND	ND	< 1	ND	6	87	209

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Conclusions:

Boeing BSS 7239 is solely a test procedure and as such, has no specific pass/fail criteria of its own. The reference criteria cited are typical for the transportation industry and are listed for reference purposes only. They may or may not apply to the specific products. Specimens are exposed to the combustion conditions described in ASTM E 662 - Standard Test Method for Specific Optical Density of Smoke Generated by Solid Materials. In compliance with this standards the material tested approve the BSS 7239 Tests

ASTM E-662 RESULTS:

REF A	Relative Room Humidity: 33% / Chamber Wall Temp: 35°C
1	Ds (1.5): 34, Ds (4.0): 52, Dm: 114.
2	Ds (1.5): 61, Ds (4.0): 89, Dm: 197.
3	Ds (1.5): 63, Ds (4.0): 91, Dm: 211.
4	Ds (1.5): 64, Ds (4.0): 94, Dm: 212.
5	Ds (1.5): 68, Ds (4.0): 97, Dm: 232.
6	Ds (1.5): 64, Ds (4.0): 95, Dm: 212.

Among the parameters normally reported are:

Ds 1.5 - specific optical density after 1.5 minutes

Ds 4.0 - specific optical density after 4.0 minutes

Dm - maximum specific optical density at any time during the 20 minute test

Authorities generally specify a maximum Ds 1.5 of 100 and a maximum Ds 4.0 of 200 in either flaming or non-flaming test mode.

Conclusions:

All specimens materials, identified in this report, when tested, meet requirements to smoke rate required by authorities.



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EN 13823 RESULTS:

Cumulative smoke production (TSP), Fire Growth Rate (FIGRA), Cumulative heat release (THR), Smoke growth rate (SMOGRA)

REF A	Lateral Flame Spread to End of Specimen	Flaming droplets	Flaming of Fallen Particle Exceeding 10s	FIGRA (w/s)	THR 600s (MJ)	SMOGRA (m²/s²)	TSP 600s (m²)
1	N	N	N	178.68	11.89	79.68	86.15
2	N	N	N	367.17	24.78	187.23	149.95
3	N	N	N	389.82	25.49	162.89	144.56
4	N	N	N	391.25	24.87	161.76	142.82
5	N	N	N	396.12	25.45	163.08	144.78
6	N	N	N	185.98	12.87	83.45	89.27

Conclusion on specimens REF A: All specimens materials, identified in this report, when tested, meet requirements to smoke rate required by authorities. (Pass)

REF B	Lateral Flame Spread to End of Specimen	Flaming droplets	Flaming of Fallen Particle Exceeding 10s	FIGRA (w/s)	THR 600s (MJ)	SMOGRA (m²/s²)	TSP 600s (m²)
1	Y	Y	Y	536.04	35.67	239.04	258.45
2	Y	Y	Y	1101.51	74.37	561.96	449.85
3	Y	Y	Y	1069.46	76.47	488.67	433.68
4	Y	Y	Y	1173.75	74.61	485.28	428.46
5	Y	Y	Y	1188.36	76.35	489.24	434.34
6	Y	Y	Y	557.94	38.61	250.35	267.81

Conclusion on specimens REF B: All specimens materials, identified in this report, when tested, do not meet smoke rate required by authorities. (Do not pass)

ISO 5659-2 RESULTS:

The radiator cone was located so that the lower rim of the radiator cone shade junction was 25 mm above the upper surface of the specimen when oriented in the horizontal position. Exposed to a radiant heat source (25kW/m² or 50kW/m²) within a closed chamber. Pilot flame was applied.

REF A	/ Relative Room Humidity: 50% / Chamber Wall Temp: 26 °C
1	Ds (1.5): 31, Ds (4.0): 49, Dm: 111.
2	Ds (1.5): 59, Ds (4.0): 86, Dm: 194.
3	Ds (1.5): 61, Ds (4.0): 88, Dm: 208.
4	Ds (1.5): 61, Ds (4.0): 91, Dm: 209.
5	Ds (1.5): 65, Ds (4.0): 94, Dm: 229.
6	Ds (1.5): 61, Ds (4.0): 25, Dm: 209.

Conclusion on specimens REF A: All specimens materials, identified in this report, when tested, meet requirements to smoke rate required by authorities. (Pass)

REF B	/ Relative Room Humidity: 50% / Chamber Wall Temp: 26 °C
1	Ds (1.5): 119, Ds (4.0): 182, Dm: 399.
2	Ds (1.5): 213, Ds (4.0): 311, Dm: 689.
3	Ds (1.5): 220, Ds (4.0): 318, Dm: 738.
4	Ds (1.5): 224, Ds (4.0): 329, Dm: 742.
5	Ds (1.5): 238, Ds (4.0): 339, Dm: 812.
6	Ds (1.5): 224, Ds (4.0): 332, Dm: 742.

Conclusion on specimens REF B: All specimens materials, identified in this report, when tested, do not meet smoke rate required by authorities. (Do not pass)



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Homologación de De Densidad de Humo NORMA IRAM 11914

Evaluación Comparativa del Índice de Densidad de Humo (Dm) de diferentes materiales, tratados con Retardante de fuego MINIT Ignífugos, según certificado de ensayo Internacional **IGS 1143-27-522** con Normativa **IRAM 11914 1**.

Comparación de Normativas, Aspecto Propósito, Método de Prueba, Criterios de Evaluación, Similitud con Normas:

ISO 5659-2 Medición de la densidad de humo en cámara cerrada. Se mide la densidad óptica de los humos en función del tiempo y calor irradiado. Densidad óptica específica máxima (Ds). Alta similitud en técnicas de medición.

EN 13823 (SBI Test) Evaluación de la densidad de humo en materiales de construcción mediante SMOGRA. Evaluación de liberación de humo durante la combustión con un objeto en llamas. Tasa de crecimiento de humo (**SMOGRA**).

Ambas normas son compatibles en parte con la evaluación de propagación del humo. IRAM 11914"

En el certificado de ensayo internacional **IGS 1143-27-522** se realizó sobre los siguientes materiales tratados con Retardante de fuego MINIT Ignífugo y no tratados en ensayos comparativos:

- Tela 100% algodón
- Tela 100% poliéster
- Alfombra 100% sintética
- Alfombra de pasto 100% sintética
- Goma espuma
- Madera

En la valuación de la densidad óptica específica corregida (Dmc) de humos en incendios se mide la absorción de luz en una cámara específica para calcular Dm. Densidad óptica específica corregida (Dmc). Especifica los mismos criterios de clasificación.

Desarrollo Comparativo de Normativas Internacionales.

ISO 5659-2 es la norma internacional utilizada para medir la densidad de humo en cámara cerrada. Su método de medición es comparable con IRAM 11914, aunque sus criterios de clasificación difieren levemente.

EN 13823 (SBI Test) evalúa la propagación del fuego y la generación de humo, midiendo SMOGRA. No es idéntico a IRAM 11914 pero se usa en conjunto para clasificar materiales.

IRAM 11914 clasifica los materiales en función de la densidad de humo corregida (Dmc):

- ≤ 132 Clasificación Baja generación de humo
- 133 – 264 Mediana generación de humo
- 265 – 396 Alta generación de humo
- > 396 Muy alta generación de humo

Para evaluar un material en Argentina según IRAM 11914, los valores obtenidos de ISO 5659-2 o EN 13823 sirven como referencia.

Conclusión Final

A partir del análisis de normativas internacionales y estudios previos, se ha determinado que los materiales tratados con retardantes de llama MINIT tienen una generación de humo moderada, lo que permite su clasificación en "**Baja a Mediana Generación de Humo**" según **IRAM 11914 (Dmc 130 - 250)**. Comparado con los materiales sin Retardante de llama.

El informe sugiere que los materiales con retardantes de llama MINIT son una opción viable para aplicaciones donde la generación de humo debe ser controlada, cumpliendo con normativas de seguridad.

Equivalencias / Homologación de ensayos aprobados comparados con normas de otros países.

Ensayos de Inflamabilidad Vertical para Textiles:

Certificaciones aprobadas: IRAM-INTI-CIT G7577, NFPA 701, & NFPA 705 Methods.

Resultado: Nivel 1 = Clase A. Resultado Cualitativo: No entra en Ignición. Pass = Complies

Equivalente con los estándares europeos:

DIN 4102 Parts 15/16 - Brandschacht (B1); NFP 92 - 503 - Bržleur flectrique;

British Standard BS 5867 and BS 5438

Ensayos de inflamafilidad horizontal para muebles y alfombras

Certificación aprobada: CAL 117 (Pass = Complies)

Equivalente con los estándares europeos:

EN 1021 - 1/2 : 1994; EN 1021 : Part 1; EN 1021 : Part 2; DIN 4102 Part 1 - Kleinbrenner (B2)
